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Case study: a well-performing hybrid plant in Hat Yai that exports without a sell-back contract

18 October 2026 · Rev05 · Prepared by Solar ESS from remote monitoring data and inverter settings. Customer details removed.

Summary

The plant is working well. Its 10 × 580 W panels (5.8 kWp) yield about 4.1 kWh per kWp per day, which is good for Hat Yai, and since the battery was added on 18 May 2026 the shop has covered 98% of its own electricity.

A check of the inverter settings on 18 Oct 2026 found two things to act on first:

1. **Export is switched on with no limit, but there is no sell-back contract.** 1,755 kWh has gone to the grid since January without payment.
2. **The grid code is set to G59/3, a UK standard,** instead of Thailand's PEA standard.

Plant at a glance

Review of all monitoring data from commissioning on 29 Jan 2026 to 17 Oct 2026 (262 days), from SolisCloud.

Item	Value
Location	Hat Yai, Songkhla
PV array	10 × 580 W panels in one string on MPPT1 = 5.8 kWp
Inverter	Solis S6-EH3P10K hybrid, 10 kW, three-phase; MPPT2 unused
Battery	One 48 V battery, added 18 May 2026
Total production	6,230 kWh (about 4.1 kWh per kWp per day)
Best day	36 kWh (25 Mar 2026), about 6.2 kWh per kWp
Total consumption	6,183 kWh (about 22–23 kWh/day)
Grid import / export	1,402 kWh / 1,755 kWh
Self-sufficiency since battery	98% (import 0.5 kWh/day, was 12.4)
Active alarms	None

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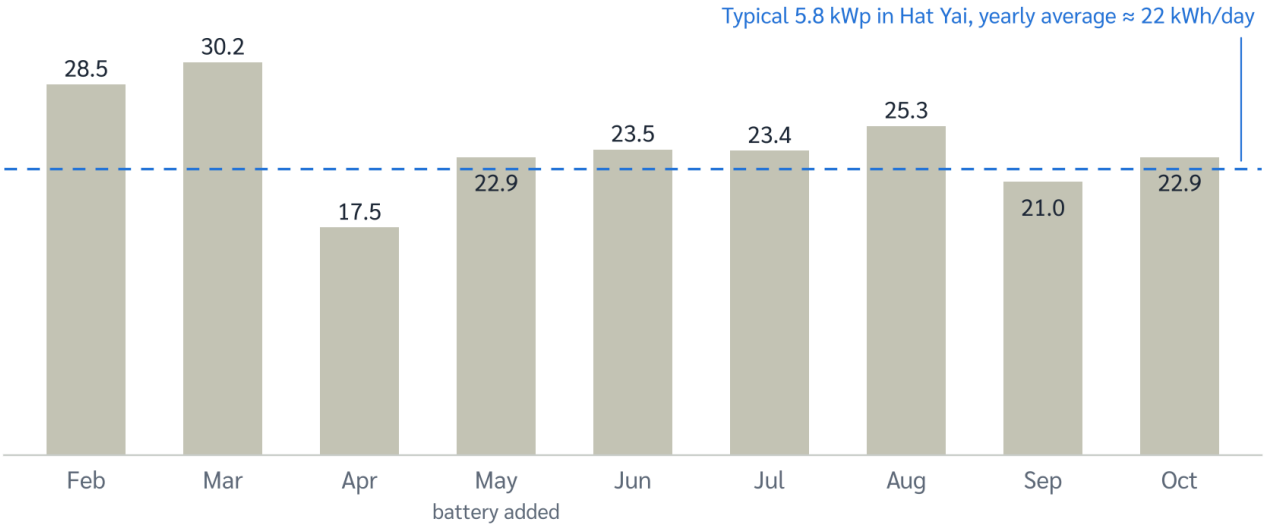
Inverter settings found on 18 Oct 2026

Setting	Value	Comment
Storage mode	Self Use	Correct for this site
Allow Export	Enable	Export is on, with no sell-back contract
Max Export Power	16,500 W	Above the inverter's 10 kW, so there is no limit
Unbalance Output	ON	Good: the shop's load sits mainly on L1 and L3
Allow Grid Charge	Enable	The battery may charge from the grid
Grid code	G59/3 (UK)	Should be the PEA grid code

Monthly performance

Production is close to or above a typical 5.8 kWp plant in most months

Average 23.9 kWh/day, Feb–Oct 2026, 109% of about 22 kWh/day for a typical 5.8 kWp plant · unit: kWh per day



Source: SolisCloud daily totals, Feb–Oct 2026 (Oct = 1–17 Oct). The typical value is an approximate yearly average.

Production is close to or above a typical 5.8 kWp plant in most months, and well above it in the dry season. On sunny days the battery is full by early afternoon.

Findings

1. Export without a sell-back contract

Export is enabled and the export limit is 16,500 W, so whatever the shop and the battery cannot use flows to the grid. Since January that is 1,755 kWh, about 3 kWh a day since the battery was added. Without a PEA sell-back contract this earns nothing. Some PEA electronic meters also record reverse flow as consumption or flag it, so it may even cost money.

2. Grid code is not PEA

The inverter runs on G59/3, a UK grid code, with that code's voltage and frequency trip limits. A grid-connected inverter in Thailand should run on the PEA grid code, and PEA normally expects grid-connected systems to be notified, even when no power is sold.

3. Surplus that could be used on site

On sunny days the battery is full by early afternoon and the rest is exported. Moving heavy loads to midday instead would use the roughly 3 kWh a day that now goes to the grid.

4. Battery energy counters do not add up

Since May the battery shows 1,832 kWh discharged but only 1,526 kWh charged. A battery cannot give out more than it takes in, so its readings are wrong. SolisCloud lists the battery maker as "OTHER". If the inverter misreads the battery's charge level, the battery can be over-discharged or never fully charged.

5. Steady overnight load

The shop draws about 0.7 kW all night, every night: about 8–9 kWh, close to 40% of daily use. Every 0.3 kW cut saves about 2.6 kWh of battery each night.

6. Inverter runs hot

The inverter's internal temperature was 58 °C at 2 kW and 58.3 °C at 3.3 kW (18 Oct 2026, 11:37 and 13:30). Inverters reduce output when they get too hot.

7. Monitoring records are incomplete

No purchase tariff is set and every kWh produced is valued at a flat 4.7 THB, so the earnings shown in the app are not meaningful. The number and wattage of the panels are not entered either.

Recommended actions

Priority	Action	Expected benefit
1	Decide the export policy: without a sell-back contract, set Allow Export to Disable (zero export); otherwise apply for PEA sell-back	Stops giving energy away and avoids meter problems
2	Have the installer change the grid code to PEA and notify PEA of the system	Meets PEA requirements
3	Check how the electricity meter records reverse flow	Avoids being billed for exported energy
4	Move heavy loads to midday, once the battery is full	About 3 kWh a day more used on site

5	Check the battery brand setting and the battery-to-inverter communication; recalibrate the charge level	Correct charge readings and longer battery life
6	List what runs overnight and switch off what isn't needed	About 2.6 kWh of battery saved per night for each 0.3 kW cut
7	Check the inverter's ventilation and shading from direct sun	Avoids output cuts on hot days
8	Enter the purchase tariff and the panel data (10 × 580 W) in SolisCloud	Correct earnings and performance figures

Notes on the data

- Source: SolisCloud daily totals since 29 Jan 2026, 5-minute inverter data for 1, 17 and 18 Oct 2026, and the inverter settings as shown in SolisCloud on 18 Oct 2026. No settings were changed.
- Array: 10 × 580 W panels, as advised. The logged string voltage (about 360 V at midday) is a little lower than usual for 10 such panels, so an open-circuit voltage check on site (about 520 V at 25 °C) is worth doing. Typical values (about 22 kWh/day for 5.8 kWp) are approximate yearly averages for southern Thailand, not a site-specific simulation.
- Prepared by Solar ESS from remote monitoring only; findings 4 and 6, and the string voltage, should be confirmed on site.